

RECYM: An Automated Multi-Feeder Workflow for Demand Allocation and Load-Flow Studies in CYMDIST Distribution Networks

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Abstract

Distribution utilities need repeatable procedures to clean network models, allocate measured substation demand, inject important-customer and prospective spot loads, and compare situational versus projected operating cases. This paper presents RECYM, a multi-feeder software suite that couples CYMDIST/CymPy with a seven-step SPA and a batch pipeline. The workflow integrates network diagnostics, large-customer energy/power mapping to SED nodes, kWh-based load allocation via the CYMDIST COM API, locked three-phase spot loads, dual load-flow scenarios, and automated delivery reports. On feeder PA217 (22.9 kV, Electro Dunas), the campaign closed with zero diagnostic problems, customers verified within tolerances, headroom-balanced allocation (9537.9 kW vs. 9537.88 kW), a 1420 kW prospective spot load, and successful dual load flows. The architecture scales to ~96 feeders sharing one Access database.

Index Terms—*CYMDIST, demand studies, distribution networks, load allocation, load flow analysis, power system modeling, software tools.*

I. INTRODUCTION

Modern distribution planning relies on detailed feeder models [1], [2]. CYMDIST provides load allocation and unbalanced load flow [3], yet campaigns often remain manual. OpenDSS [5] and pandapower [6] support research automation, while many Latin American utilities standardize on .zxst/.mdb assets. RECYM bridges that gap [7], [8] with a layered architecture, a §§1–7 campaign, and validation on PA217 (22.9 kV).

II. RELATED WORK

Kersting [1] formalizes three-phase modeling. Allocation uses energy/demand/kVA weights [3], [9]. COM automation of commercial tools is critical when the system of record is proprietary [7], [8]. Capacity and interconnection studies motivate situational vs. projected narratives [12]–[14].

III. SYSTEM ARCHITECTURE

Five layers: presentation (React SPA), API (FastAPI jobs/SSE + Flask bridge), domain (pipeline/analysis/optimization), CymPy/COM core, and external data (.zxst/.mdb/Excel). Multi-feeder config via settings.json + feeders/<ID>.json (~96 feeders + ELD study).

Fig. 1. RECYM layered architecture



Fig. 1. RECYM layered architecture (UI → API → domain → CYMDIST core → data).

IV. MATHEMATICAL MODELS AND ALGORITHMS

Model 1 (kWh allocation): $P_{cab_res} = P_{cab} - \sum P_{fix}$; $P_i = P_{cab_res} * E_i / \sum E_u$; balance error $\epsilon_{ps_P} = |\sum P_i - P_{cab}| / P_{cab} * 100\%$. Precision MAE/RMSE of DeltaP, DeltaE (tol. 0.01 kW / 0.5 kWh).

Model 2 (SpotLoad + dual LF): $P_{phase} = P_{3ph}/3$, $Q_{phase} = Q_{3ph}/3$; $y_{sit} = F(x; c=0)$, $y_{prj} = F(x; c=1)$; $\Delta y = y_{prj} - y_{sit}$; $\delta_{P_P} = 100 * \Delta P / P_{sit}$. Pseudocode Algorithms 1-3. Sample PA217 validation: $\epsilon_{ps_P} \sim 7.6e-5\%$, MAE=0, $\delta_{P_P}=+17.17\%$, closure 21/21.

IV-bis. OPERATIONAL WORKFLOW

IV. PROPOSED WORKFLOW

§1 context/headroom; §2 quality gate; §3 customers EA→KWH / Pot→kW Locked + allocation; §4 SpotLoad P/3 Q/3 Locked (no re-allocation afterward); §5 situational (disconnect §4) / projected (connect §4) LF; §§6–7 reports and suite.

Fig. 2. RECYM campaign workflow §§1–7

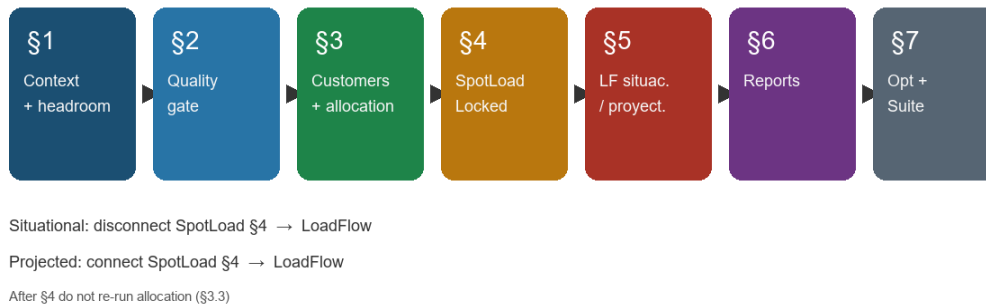


Fig. 2. RECYM operational campaign §§1–7.

V. CASE STUDY

RECYM is already wired to the Electro Dunas Access database with 97 feeder configurations and multiple .zxst studies; PA217 (22.9 kV) is reported as a representative experimental sample. On PA217: zero diagnostic problems, verified customers, headroom-balanced allocation (9537.9 vs 9537.88 kW), 1420 kW SpotLoad, and dual LF with quantified P/Q/S deltas.

Corpus: 97 feeders + ELD; shared DB; multiple .zxst. PA217 is a sample to validate RECYM (not the delivery report in doc/).

VI. RESULTS AND DISCUSSION

Diagnostics: 0 messages / 0 problems. Customer precision: 11 OK, 0 fails (Table I). COM allocation: $\Sigma P=9537.9$ kW vs headroom 9537.88 kW ($\Delta \approx 0\%$), 257 loads. Locked SpotLoad 1420 kW at NODE_1080_320357_MI-17145_2 (Fig. 3). Situational and projected LF OK (Figs. 4–5). Automated closure 21/21 steps.

TABLE I. IMPORTANT-CUSTOMER PRECISION ON PA217 (SAMPLE)

Customer	SED	EA (kWh)	Pot (kW)	ΔkWh	ΔkW	Status
Caliza cementos inca	SE30940	1,747,551.27	3800.57	0.0	0.0	OK
Minerales Paracas	SE30480	678,852.12	1273.76	0.0	0.0	OK
LJMetales	SE30642	527,079.81	956.49	0.0	0.0	OK
Emsalsa	SE30353	157,621.44	458.00	0.0	0.0	OK
Uvica	SE30424	167,101.44	366.30	0.0	0.0	OK



Fig. 3. Location of the prospective SpotLoad (1420 kW) on PA217.

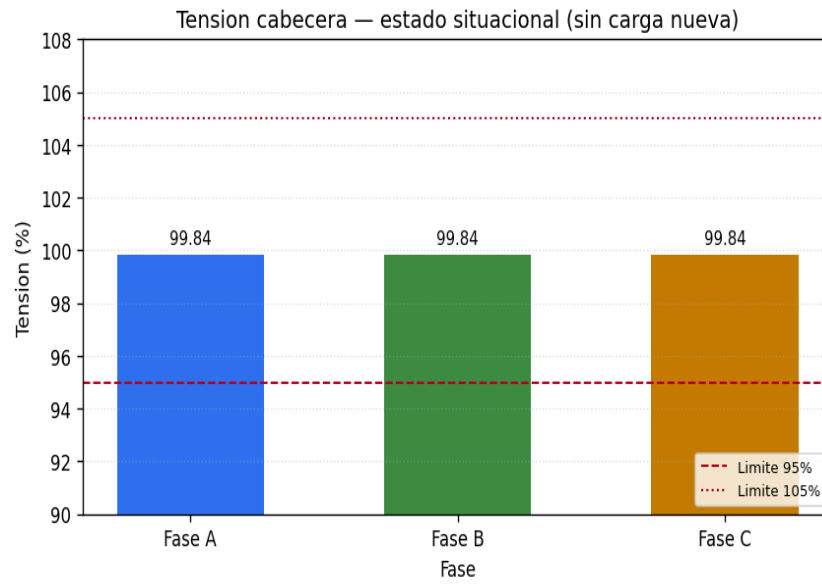


Fig. 4a. Voltage map — situational scenario (SpotLoad §4 disconnected).

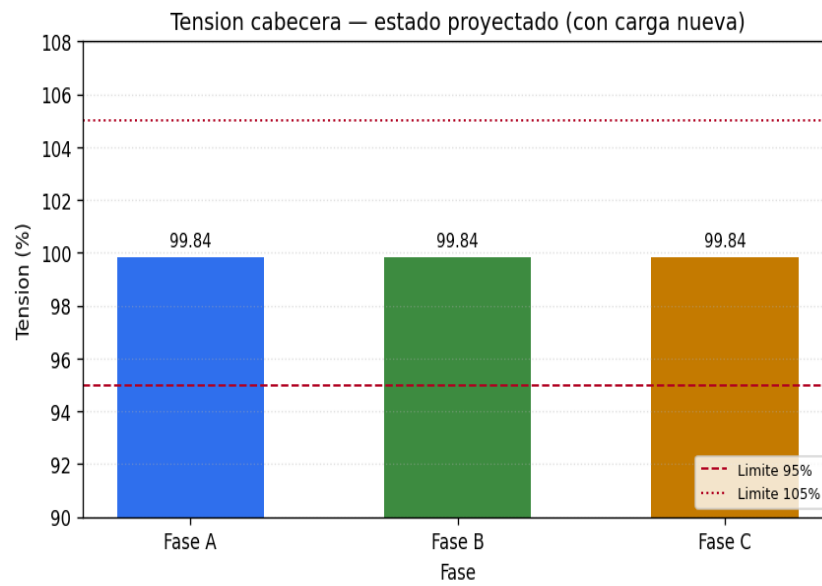


Fig. 4b. Voltage map — projected scenario (SpotLoad §4 connected).

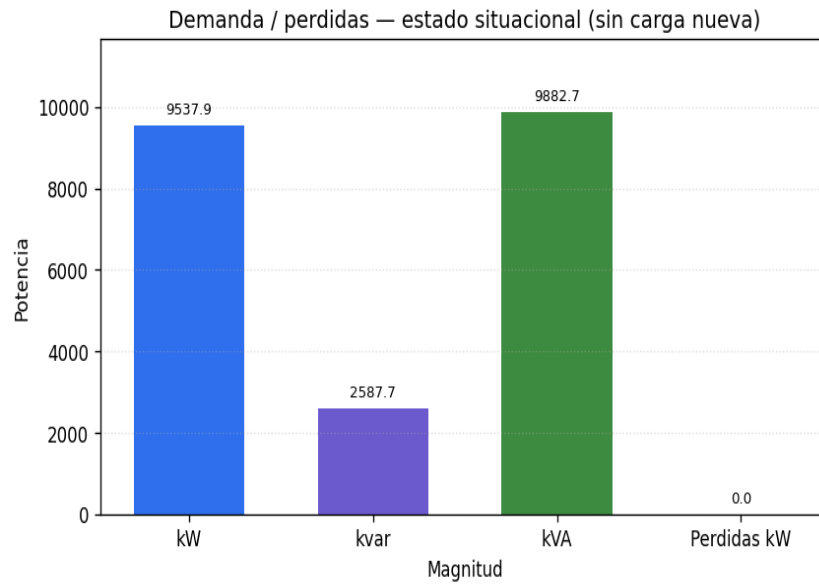


Fig. 5a. Loading — situational scenario.

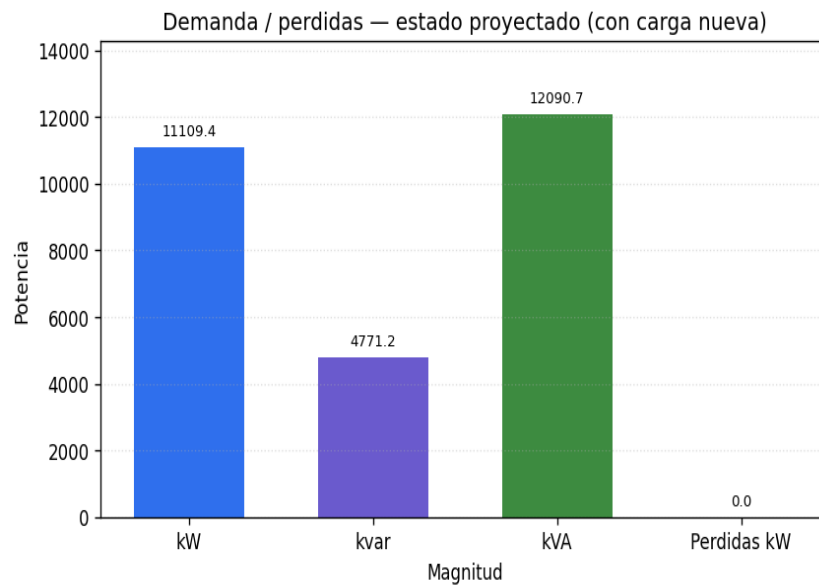
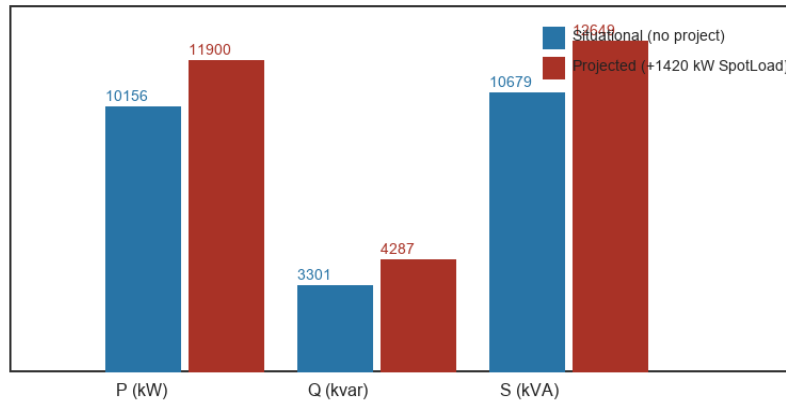


Fig. 5b. Loading — projected scenario (sample PA217).

Fig. 6. PA217 load-flow demand: situational vs projected



Source: PA217 COM LoadFlow narrative (Electro Dunas / RECYM)

Fig. 6. Headroom demand situational vs projected (real COM LF data, PA217).

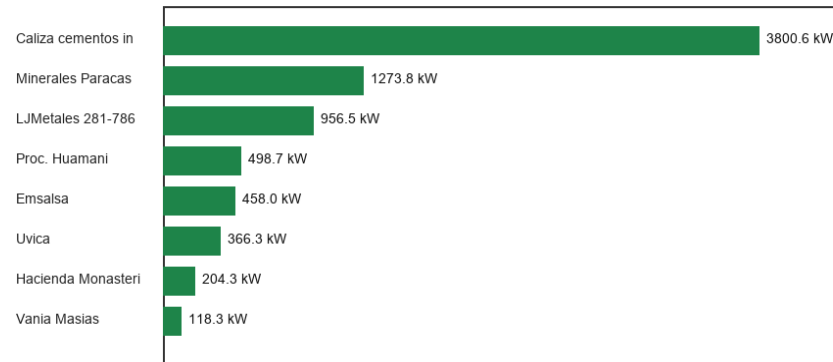
Fig. 7. PA217 headroom voltage and power factor

Metric	Situational	Projected	Delta
Vhead A/B/C (%)	99.84 / 99.84 / 99.84	99.84 / 99.84 / 99.84	0.00 pp
Voltage drop (pp)	0.16	0.16	0.00
Power factor (%)	95.10	94.08	-1.02
Head P (kW)	10156.0	11900.0	+1744.0
Head Q (kvar)	3301.0	4287.0	+986.0
Head S (kVA)	10679.0	12648.7	+1969.7

Source: data/output/feeders/PA217/demand/informe_narrative.txt

Fig. 7. Visual Table III: voltage, PF and P/Q/S (sample PA217).

Fig. 8. Verified important-customer Pot (kW) on PA217



Source: precision_report.json (tol 0.5 kWh / 0.01 kW); n_ok=11 n_fail=0

Fig. 8. Verified important-customer Pot (kW) on sample PA217.

Fig. 9. End-to-end campaign closure on PA217 (21/21 OK)

1.0 archivos	n_feeder=96
1.1 aplicar	Contexto OK: PA217: red NET_2030_179_PA217: Tablero 2 reiniciado (ejec
1.2 cabecera	P=9537.88 kW Q=2587.65: NET_2030_179_PA217
2.0 estado	ready=False converge=None
2.1 diagnosticar	Diagnóstico OK: 0 msgs: problemas=0: tablero actualizado
2.2 tablero	regenerado
3.0 archivos	suministro=ML_1225.xlsx: CI=0226clientesImportantes.xlsb
3.1 tabla	filas=12 (reintento con clientes_file)
3.2 aplicar EA/Pot	EA: Consumo(KWH) + Pot: kW Locked: 253 OK: 0 excluidas: 0 sin SED: K
3.3 distribución	fallback_KWH_api: total=18.25s: Validación OK: Suma kW=9537.9 vs cabecer
4.1 inventario nodos/cargas	n=?
4.2 spotload nueva	omitido (ya existe last_spot_load en sesión)
5.1 situacional	ok
5.2 proyectado	ok
6.1 armar informe	armar OK tras meta OCR
6.2 status entrega	delivery_ready=False missing=informe_meta_ocr (cliente + potencia_kw via PDF)
7.1 entorno	ok
7.2 conexión	Conexión Electro Dunas OK
7.3 validar entradas	Validación ejecutada
7.4 pipeline	PIPELINE OK
6.2b delivery	delivery_ready=True missing=[]

closed=True n_ok=21 n_fail=0 | cierre_1_7_report.json

Fig. 9. Automated §§1–7 closure on PA217 (21/21 OK).

Limitations: Windows/COM; study integrity; COM fallback. Detailed tables are for sample PA217; the catalog already has 97 feeders and multiple studies—fleet KPIs remain future work.

VII. CONCLUSION

RECYM automates multi-feeder CYMDIST campaigns across 97 Electro Dunas feeders and multiple .zxst studies. With PA217 as representative sample: zero diagnostic issues, verified precision, balanced allocation, dual LF for 1420 kW, and 21/21 closure. Future work: fleet KPIs over the full catalog and §7 hardening.

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